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Abnormality classification from electrocardiograms with various lead combinations

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E-mail: xieguotong@pingan.com.cn and yichongli.cvd@139.com**Keywords:** reduced leads, deep neural network, multi-label ECG classification, model generalization

Abstract

Objective. As cardiovascular diseases are a leading cause of death, early and accurate diagnosis of cardiac abnormalities for a lower cost becomes particularly important. Given electrocardiogram (ECG) datasets from multiple sources, there exist many challenges to the development of generalized models that can identify multiple types of cardiac abnormalities from both 12-lead ECG signals and reduced-lead ECG signals. In this study, our objective is to build robust models that can accurately classify 30 types of abnormalities from various lead combinations of ECG signals. **Approach.** Given the challenges of this problem, we propose a framework for building robust models for ECG signal classification. Firstly, a preprocessing workflow is adopted for each ECG dataset to mitigate the problem of data divergence. Secondly, to capture the lead-wise relations, we use a squeeze-and-excitation deep residual network as our base model. Thirdly, we propose a cross-relabeling strategy and apply the sign-augmented loss function to tackle the corrupted labels in the data. Furthermore, we utilize a pos-if-any-pos ensemble strategy and a dataset-wise cross-evaluation strategy to handle the uncertainty of the data distribution in the application. **Main results.** In the Physionet/Computing in Cardiology Challenge 2021, our approach achieved the challenge metric scores of 0.57, 0.59, 0.59, 0.58, 0.57 on 12-, 6-, 4-, 3- and 2-lead versions and an averaged challenge metric score of 0.58 over all the lead versions. **Significance.** Using the proposed framework, we have developed the models from several large datasets with sufficiently labeled abnormalities. Our models are able to identify 30 ECG abnormalities accurately based on various lead combinations of ECG signals. The performance on hidden test data demonstrates the effectiveness of the proposed approaches.

1. Introduction

Cardiovascular diseases are chronic non-communicable diseases that gradually evolve throughout life. They have been the leading cause of premature mortality and disability for years across the world. It is estimated that by 2030, 23.6 million people will die from cardiovascular diseases each year (Francula-Zaninovic and Nola 2018), making this a research topic of great importance in healthcare. There are various classes of cardiac abnormalities, and it is crucial to diagnose cardiac abnormalities as early as possible because they can cause the most serious life-threatening diseases (Amara *et al* 2017).

The multi-lead electrocardiogram (ECG) is a powerful tool for describing the activity of the heart; it is a safe and painless test used to quickly detect heart problems and monitor the heart's health status. At present, many studies have reported automated analysis of ECG signals with time-frequency transformation (Alexakis *et al* 2003, Mahmoodabadi *et al* 2005), machine learning techniques (Vafaie *et al* 2014, Chen *et al* 2018) and deep neural networks (Natarajan *et al* 2020, Zhu *et al* 2021). Although the 12-lead ECG is the standard diagnostic

screening system for many cardiological issues (Kligfield 2002), the limited accessibility of 12-lead ECG devices provides a rationale for using smaller, lower-cost devices such as portable or handheld ECG devices (Samol et al 2019). These convenient devices usually record reduced-lead signals. Besides, the ECG signal is easily affected by the wrong lead placement and patient positioning, in which case some leads are unavailable. Thus, the study of classification of ECG abnormalities on reduced ECG leads is meaningful. Researchers have found evidence that subsets of the standard 12 leads can capture useful information (Aldrich et al 1987, Drew et al 2002, Green et al 2007), and may even be comparable to 12-lead ECG in some limited contexts.

In this work, we attempt to develop robust deep learning models to identify 30 types of ECG abnormalities extensively from various lead combinations. The use of multi-source ECG datasets to develop well-generalized models faces four challenges: (1) *Diversified signal characteristics*: usually, ECG signals may differ in sampling frequency, length, noise level and scale. Especially for signals from different data sources, the divergence of these characteristics will become much more significant because of different device specifications. (2) *Unknown lead-wise relations*: in ECG classification, the importance of each lead varies with the change of input signals and lead versions because the features of different abnormalities are reflected on different leads. For better generalization in practical applications, it is significant to develop extendable models that could correctly capture these lead-wise relations. (3) *Corrupted labels in training data*: ECG signals are usually partially labeled with only those abnormalities considered in real-world ECG datasets. Previous work (Zhu et al 2021) discussed how these partial labels confuse model learning. Some abnormalities are ignored in specific data sources. Even within a single dataset, the labels of some abnormalities could be missing for specific samples, although these abnormalities are correctly labeled for other signals. (4) *Unknown data distribution in applications*: although we could develop ECG classification models from multi-source ECG datasets, we never know how the data are distributed in the real-world application of abnormality identification. Data shift is a general problem in machine learning and deep learning (Quiñonero-Candela et al 2009). In the ECG classification domain this becomes more serious because of the existence of the three other challenges mentioned above. Considering the uncertainty of data distribution, we need to design model-ensembling methods and to evaluate the generalizability with only training data.

To address these challenges, we propose an extendable framework for identification of ECG abnormalities. The contributions of this work are summarized as follows: (1) We design a series of approaches for addressing the summarized challenges in ECG classification with deep learning methods. (2) We use the proposed framework to build the models based on several large ECG datasets from different countries with sufficient labels. The models can identify 30 types of ECG abnormalities and handle various lead combinations. (3) we conduct both offline and online experiments to examine the performance of the proposed approaches. The results show the good generalizability of our models and demonstrate the effectiveness of the proposed approaches. The updates we made to our previous solution and the comparisons with other leading solutions are presented in the Discussion section.

2. Methods

In this section, we introduce our proposed methods for identifying 30 ECG abnormalities from multi-source ECG databases with various lead versions. We first introduce the preliminaries of the PhysioNet/Computing in Cardiology Challenge 2021 (Reyna et al 2021, 2022), which extended the 2020 Challenge (Alday et al 2020, Goldberger et al 2000). Secondly, we describe our data-preprocessing techniques to reduce signal diversity. Thirdly, we present our basement model that we use to capture lead-wise relations. Then, we introduce the cross-relabeling strategy and sign-augmented loss function, which are used to relieve the corruption of the labeling system. Finally, we propose our ensemble and evaluation strategy to handle the uncertainty during testing. The combined framework of proposed approaches is shown in figure 1.

2.1. Preliminaries

The PhysioNet/Computing in Cardiology Challenge 2021 provided multi-source 12-lead ECG datasets and defined the metric for identification of ECG abnormalities. Challengers were expected to develop models that could accurately identify ECG abnormalities from ECG signals with various lead combinations (see table 1).

2.1.1. Datasets

Organizers made some of the 12-lead challenge data publicly available as development data and kept the remaining data hidden as official validation data and official test data with different lead combinations. In the development data, there are over 88 000 ECG recordings with labeled abnormalities from eight datasets, i.e. the China Physiological Signal Challenge 2018 (CPSC and CPSC_Extra) database, the Physikalisch-Technische Bundesanstalt (PTB and PTB_XL) database, the Georgia 12-lead ECG Challenge (Georgia) database, the Chapman University, Shaoxing Peoples Hospital (Chapman) database, the Ningbo First Hospital (Ningbo)

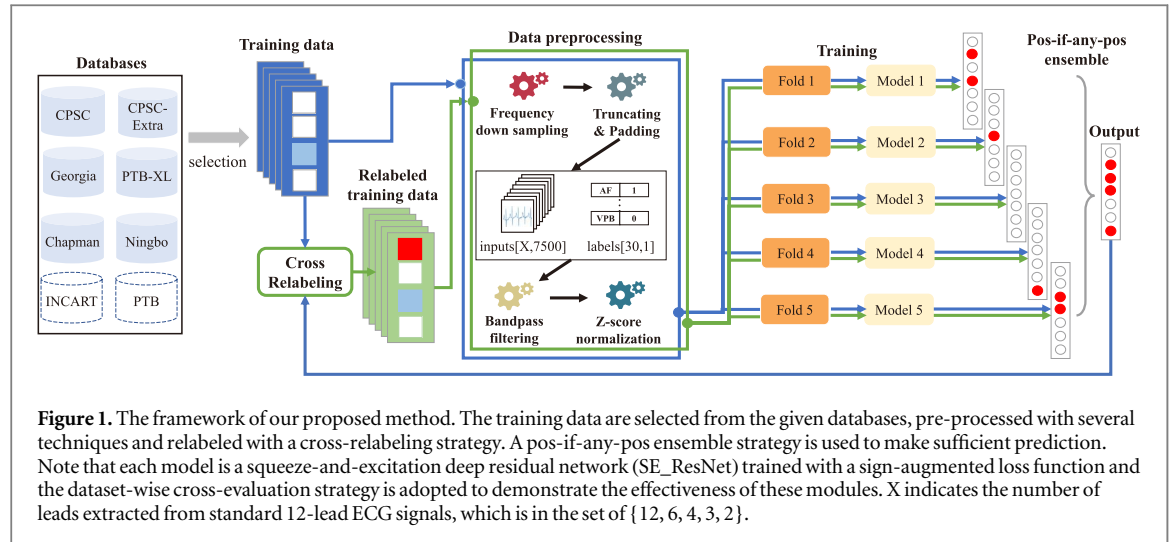


Table 1. Lead combinations used in the 2021 PhysioNet/Computing in Cardiology Challenge.

Number of leads	Lead combination
12	I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6
6	I, II, III, aVR, aVL, aVF
4	I, II, III, V2
3	I, II, V2
2	I, II

Table 2. Statistics of the development data. Recordings with scored labels are used as training data except datasets marked with star.

Dataset	Recordings	Recordings with scored labels
CPSC	6877	5279
CPSC-Extra	3453	1278
*INCART	74	34
*PTB	516	116
PTB_XL	21 837	21 604
Georgia	10 344	9458
Chapman	10 247	9710
Ningbo	34 905	34 487

database and the St Petersburg Institute of Cardiological Technics 12-lead Arrhythmia (INCART) database. Additional CPSC and Georgia databases were kept hidden as official validation data and part of the official test data. An undisclosed American database (Undisclosed) and a new database from the University of Michigan (UMich) were also kept hidden as part of the official test data. In our experiments, we excluded development recordings without scored labels and then removed INCART and PTB databases because of sparse label distribution (see table 2). As a result, there are 81 816 recordings in our training data.

2.1.2. Metrics

Organizers defined a scoring metric that awards partial credit to a misdiagnosis that results in similar treatments or outcomes as the ground truth diagnosis. Let $C = [c_i]$ be a collection of all scored classes and $A = [a_{ij}]$ be a computed multi-label confusion matrix, where a_{ij} is the number of recordings in a database that were predicted as belonging to class c_i but actually belong to class c_j . a_{ij} is calculated by $a_{ij} = \sum_{n=1}^N a_{ijn}$, where N is the number of ECG recordings and a_{ijn} is defined by

$$a_{ijn} = \begin{cases} \frac{1}{|x_n \cup y_n|}, & \text{if } c_i \in x_n \text{ and } c_j \in y_n. \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

x_n is the set of positive labels and y_n is the collection of positive predictions for the n th recording. The weight matrix $W = [w_{ij}]$ is defined by organizers to measure the similarities among abnormalities, where w_{ij} denotes the reward for a positive classifier output for class c_i with a positive label c_j . The un-normalized score is then calculated by $s_{\text{unnorm}} = \sum_i \sum_j w_{ij} a_{ij}$. The final (normalized) challenge metric score is calculated as

$$s_{\text{norm}} = \frac{s_{\text{unnorm}} - s_{\text{inter}}}{s_{\text{correct}} - s_{\text{inter}}} \quad (2)$$

where s_{correct} is the score for a classifier that always predicts the true label and s_{inter} is the score for an inactive classifier that always outputs a normal class. The higher the score indicates the better performance of the model.

2.2. Data preprocessing

The raw data from different data sources vary in signal length, sampling frequency, signal amplitude and noise level. To remove this dataset-specific divergence that would reduce the generalizability of the classification model we adopted the preprocessing techniques detailed in the following.

2.2.1. Resampling, truncating and padding

During training and testing, signals were re-sampled to a unified sample frequency (500 Hz). To address the variable length of all training ECG recordings, we randomly extracted 15 s of a recording if the recording was longer than 15 s, i.e. 7500 points for 500 Hz. If the length of a recording was shorter than 15 s zero-padding was applied to extend the signals to 15 s.

2.2.2. Noise reduction and amplitude normalization

Given the importance of filtering both the high-frequency and low-frequency information reported by Nejedly *et al* (2021), in this work, we adopted a Butterworth bandpass filter to reduce the noise with a frequency bandwidth from 1 Hz to 47 Hz. For each lead of each ECG recording, the amplitude was normalized in the same way so that the training and testing data shared similar amplitudes. We chose Z-score normalization, as it could prevent the baseline position from deviating too much from a zero value. Then, the signals were multiplied 100 times.

2.3. Base model

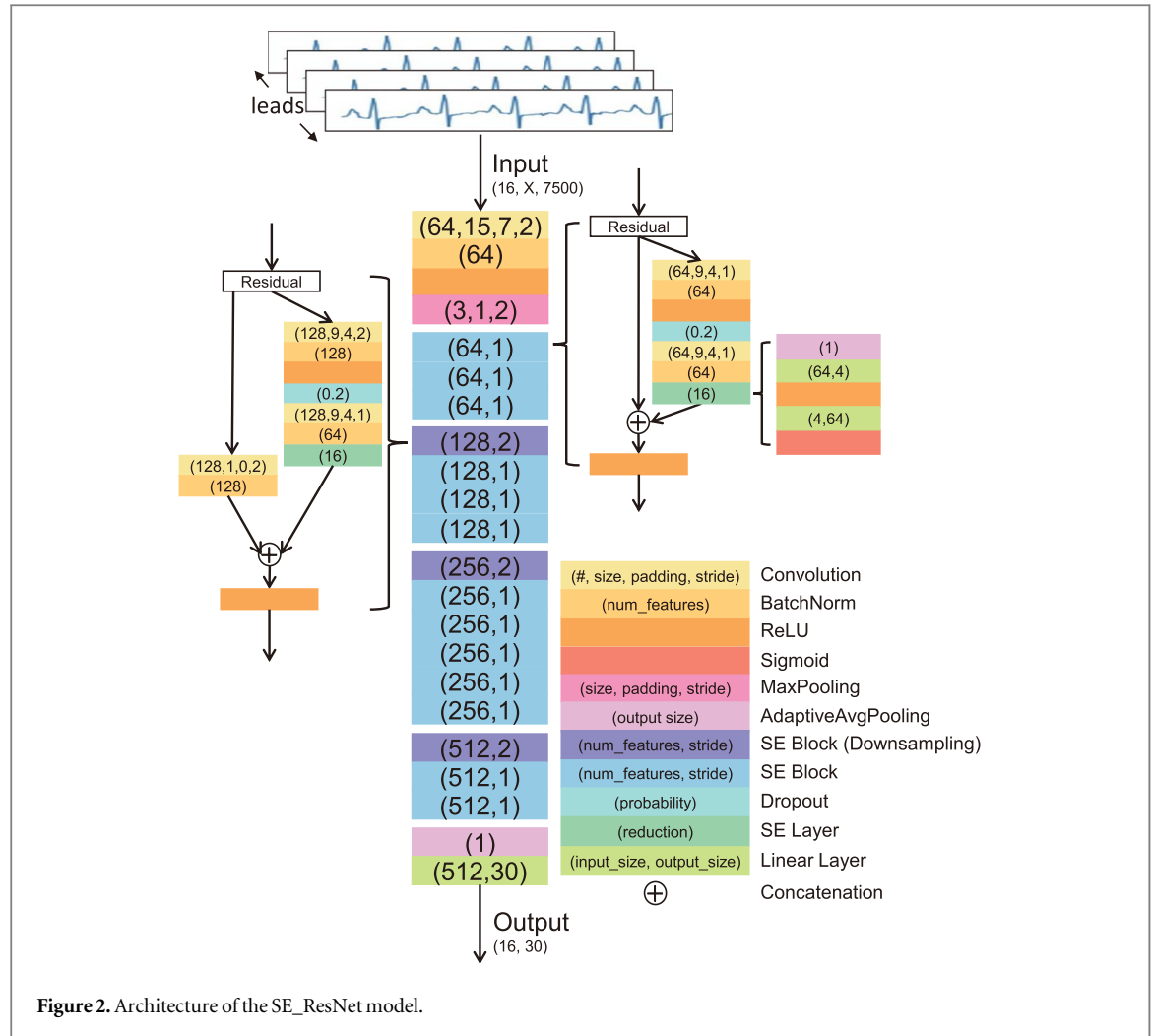
In order to capture lead-wise relations for various input signals and lead versions, we used the SE_ResNet (Hu *et al* 2018) model shown in figure 2 as our primary model. The base model consists of one convolutional layer followed by several residual blocks, each of which contains two convolutional layers and a squeeze-and-excitation (SE) block. Each SE block adopts a squeeze operation and an excitation operation to the input from the last layer. For each channel, the squeeze operation compresses the global spatial information with global average pooling and produces an embedding of the global distribution. Thus, all the layers could fetch the information from the global field. The excitation operation takes this embedding as input, captures the channel-wise dependences and then outputs weights for each channel. Consequently, these weights are applied to previously learned feature maps to measure the feature importance and realize the feature recalibration. To distinguish wider features, we used a large kernel size in all the convolutional options.

2.4. Missing label relieving

There were six datasets used for training, where each dataset is partially labeled and none contain all 30 abnormalities. For example, as shown in table 7, complete left bundle branch block and poor R wave progression only appear in the Ningbo dataset. Also, for a specific sample, some abnormalities could be missing labels because of annotation error. To handle the missing label problem, we used a relabeling strategy and an advanced loss function to mitigate the dataset-wise global missing and sample-wise local missing labels, respectively.

2.4.1. Cross-relabeling

To help the model correctly understand the abnormality features and reduce misjudgment, previous works revised confused labels with manual relabeling (Clifford *et al* 2017, Teijeiro *et al* 2017). However, manual relabeling is not applicable for large sample sizes and is limited by different annotation propensities. Some works (Nonaka and Seita 2020, Zhu *et al* 2021) applied relabeling models that were derived from the same distribution as the samples to be relabeled, where consistent distribution will restrict the relabeling accuracy. In our approach, a cross-relabeling strategy was adopted to mitigate the global dataset-wise missing labels, where the relabeling models were derived from a different distribution from the samples to be relabeled. For each dataset, several SE_ResNet models were trained with the other five datasets and then used for inferring pseudo labels on globally missing-labeled classes of this dataset. Specifically, for each ECG signal, inferred pseudo labels were



regarded as true labels if (1) the inference probability was higher than a relabeling threshold τ and (2) the labels were generally missing labels in the corresponding dataset.

2.4.2. Sign-augmented loss

Usually, the binary cross entropy (BCE) loss function is a common solution in multi-label classification. However, the existence of local missing labels in each sample would confuse the model learning as some potential abnormalities were ignored. Inspired by Sun *et al* (2019), we utilized the difference between predicted probability and the ground truth to modify the weight of the BCE loss function. Specially, we proposed an extensible sign-augmented module to overcome the local missing labels by reducing the accumulated loss from negative labels and thus focusing more on positive labels as the number of the negatives was much larger than the positives. Formally, for each class, the sign-augmented loss (SAL) is defined as

$$\text{SAL} = \begin{cases} (y - p)^{\gamma_+} \times \text{BCE}(p, y), & \text{if } |y - p| < 0.5 \\ (y - p)^{\gamma_-} \times \text{BCE}(p, y), & \text{otherwise} \end{cases} \quad (3)$$

where y is the ground truth, p is the output probability and the sign difference is enhanced by parameters $\gamma_+ > \gamma_- \geq 0$. Note that the positive augmentation parameter γ_+ and the negative augmentation parameter γ_- are both even numbers. By using SAL, the accumulated loss from the large amount of true negatives (where many samples are missing labels) becomes much smaller with the unbalanced augmentation, and the gradient of the loss function has a smoother change around low and high probability but a sharper change around 0.5 probability.

2.5. Towards unknown distribution in application

Given the significant distributional difference among the six considered training datasets, unseen distribution in applications is likely to be greatly different from all the six training distributions. To handle this kind of test uncertainty, we adopted an ensemble strategy to make sufficient prediction and an offline evaluation strategy to evaluate the model performance on unseen distributions.

Table 3. Performance of 12-lead models on dataset-wise cross-evaluation. All the models were developed based on our baseline model. Each score is the challenge metric score on an offline validation set that was left out during dataset-wise cross-evaluation. From this table we select Model 5 as our best model. *T1* denotes bandpass filtering and *Z*-score normalization, *T2* denotes SAL, *T3* denotes the pos-if-any-pos ensemble and *T4* denotes cross-relabeling. *D1* to *D6* denote leaving out CPSC, CPSC_Extra, PTB_XL, Georgia, Chapman and Ningbo, respectively, as the offline validation set in dataset-wise cross-evaluation.

Model	<i>T1</i>	<i>T2</i>	<i>T3</i>	<i>T4</i>	<i>D1</i>	<i>D2</i>	<i>D3</i>	<i>D4</i>	<i>D5</i>	<i>D6</i>	Average validation score
Model 1	×	×	×	×	0.45	0.50	0.04	0.36	0.71	0.62	0.45
Model 2	✓	×	×	×	0.41	0.47	0.07	0.50	0.71	0.62	0.46
Model 3	✓	✓	×	×	0.44	0.49	0.07	0.50	0.71	0.62	0.47
Model 4	✓	✓	✓	×	0.41	0.49	0.15	0.52	0.71	0.64	0.49
Model 5	✓	✓	✓	✓	0.41	0.51	0.16	0.54	0.77	0.69	0.51

2.5.1. Pos-if-any-pos ensemble

Similar to our previous solution (Xu et al 2021), we conducted K -fold cross-validation on a training set to build K classifiers (SE_ResNet models), where each classifier learned from $(K - 1)/K$ of the shuffled training data. To make an overall sufficient prediction on an unseen distribution, a pos-if-any-pos ensemble strategy was used to identify abnormalities as much as possible:

$$y_n = \bigcup_{k=1}^K y_n^k \quad (4)$$

where y_n^k is the set of positive predictions of the k th classifier for the n th recording. With this strategy, a class was predicted to be positive if any of the K classifiers in the ensemble predicted the class to be positive.

2.5.2. Dataset-wise cross-evaluation

To evaluate the generalization performance on the distribution of interest, previous works tried to separate the distribution of interest from the given data as an offline validation set (Han et al 2021, Li et al 2021). Inspired by Han et al (2021), we adopted a dataset-wise cross-evaluation approach. For the six datasets considered in our training data, we used five to develop the model and the other as the offline validation set to validate the model performance. This process was repeated six times until each dataset was left out to validate the performance. Finally, we report the averaged offline validation performance over the six-times cross-evaluation. Dataset-wise cross-evaluation has the potential to validate the model performance on a new distribution.

2.6. Training setup

For each lead combination in table 1, we performed the proposed approaches to develop $K = 5$ SE_ResNet models with a kernel size of 9 for the ensemble. Note that the cross-relabeling strategy was applied only once with 12-lead models and the inferred pseudo labels were used in the model training of all lead combinations. Each of the SE_ResNet models was trained for several epochs (no more than 13) with batch size of 32 until the validation loss showed no further decrease. During training, the learning rate was 0.001 at the beginning and was rescheduled to 0.0001 at the 10th epoch. The relabeling threshold τ was set to be 0.8 to ensure high confidence in relabeling. The positive augmentation parameter γ_+ was set as 2 and the negative augmentation parameter γ_- was set as 0. A single threshold of 0.35 was set to binarize the outputs of all the models.

3. Results

3.1. Offline validation results

We conducted an ablation experiment with dataset-wise cross-evaluation to evaluate the performance of each proposed module. Table 3 shows the offline performance of the 12-lead models. Model 1 is a baseline model that uses SE_ResNet with frequency resampling and length normalization. In Model 2, we apply bandpass filtering and *Z*-score normalization. Compared with the baseline model, the performance of Model 2 improved by further reducing the diversity of signal properties. We utilize SAL in Model 3 over Model 2 to relieve the sample-wise missing labels. The performance of Model 3 demonstrates that handling the local missing labels helps to improve the generalizability on unseen distributions in application. To make sufficient prediction, the pos-if-any-pos ensemble strategy is adopted in Model 4, obtaining a much higher averaged score. Finally, Model 5 takes pseudo labels inferred with cross-relabeling into account and achieves a average score of 0.51, which is the best score. We therefore select Model 5 as our best model.

Table 4. Performance comparison of our best model (HeartBeats) with other state-of-the-art solutions. Each score is the challenge metric score on the online testing dataset. We also include the average score which is computed as the mean of the 12-, 6-, 4-, 3- and 2-lead scores. The best performances are marked in bold.

Method	12-lead	6-lead	4-lead	3-lead	2-lead	Average
ISIBrno-AIMT	0.58	0.58	0.58	0.58	0.58	0.58
NIMA	0.55	0.58	0.58	0.57	0.57	0.57
DSAIL_SNU	0.55	0.52	0.56	0.55	0.56	0.55
HeartBeats	0.57	0.59	0.59	0.58	0.57	0.58

Table 5. Challenge metric scores of our best model on different data sources in the online testing dataset.

Data source	12-lead	6-lead	4-lead	3-lead	2-lead
CPSC	0.74	0.65	0.64	0.55	0.55
Georgia	0.63	0.60	0.62	0.60	0.60
Undisclosed	0.52	0.56	0.54	0.56	0.53
UMich	0.57	0.59	0.60	0.58	0.58

3.2. Online testing results

In table 4, we compare the challenge metric scores of our best model (named HeartBeats) with other state-of-the-art solutions on an online testing dataset for different lead versions. Specifically, we compare our solution with ISIBrno-AIMT (Nejedly *et al* 2021), NIMA (Han *et al* 2021) and DSAIL_SNU (Wickramasinghe and Athif 2021), which are also the leading solutions in the Physionet/Computing in Cardiology Challenge 2021. Our method achieved the challenge metric scores of 0.57, 0.59, 0.59, 0.58, 0.57 on 12-, 6-, 4-, 3- and 2-lead combinations, respectively, and achieved the best performance on 6-, 4- and 3-lead versions and the averaged score. The online results demonstrated our model's ability to classify the ECG abnormalities for various lead versions despite the online test distribution not being seen during training. The detailed performances on different data sources are shown in table 5.

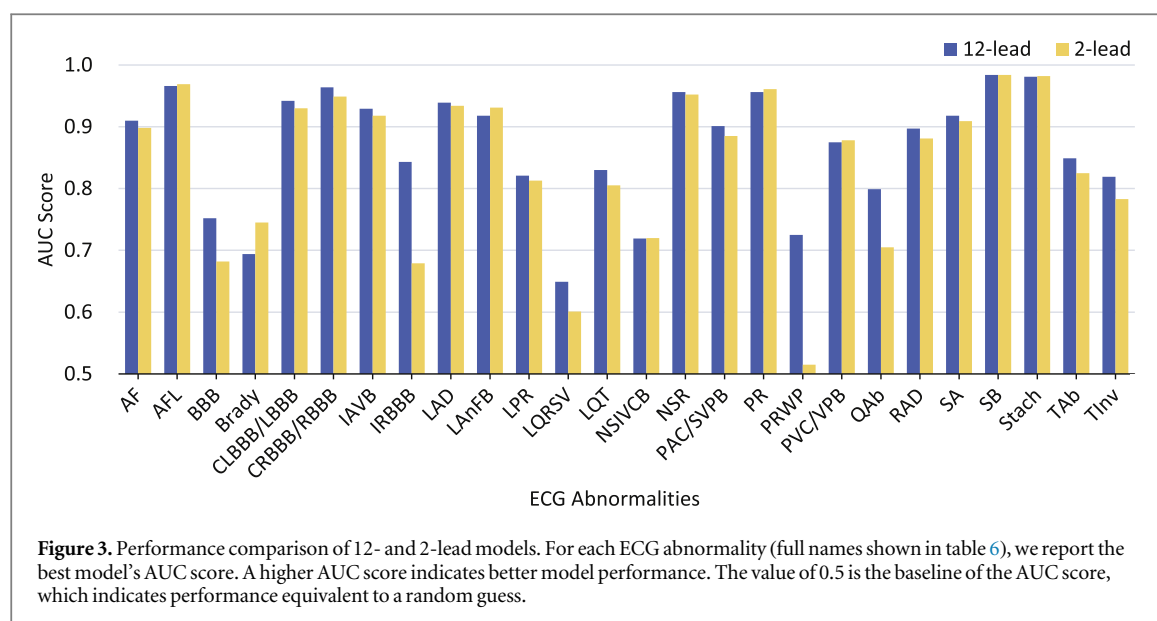
3.3. Detailed analysis

For the 30 abnormalities considered, four pairs were treated as the same ECG abnormality when the organizers of the PhysioNet/Computing in Cardiology Challenge calculated the score (complete left bundle branch block and left bundle branch block, complete right bundle branch block and right bundle branch block, premature atrial contraction and supraventricular premature beats, premature ventricular contraction and ventricular premature beats). Based on this, we analyze the area under the curve (AUC) scores for all the 26 categories on 12-lead and 2-lead models (Abberavation of scored disease names is shown in table 6).

As shown in figure 3, for most of the categories identification of ECG abnormalities with 2-lead data could achieve similar performance to 12-lead data. For some categories, performance of 2-lead models shows a significant drop from 12-lead models, which indicates that the missing signals from 10 leads contribute additional information for these categories. Poor R wave progression in particular can hardly be judged from 2-lead signals. A special case is bradycardia: 2-lead models improve on this category over 12-lead models. The most probable reason is that bradycardia is seriously mislabeled in PTB_XL, Georgia, Chapman and Ningbo datasets (as shown in table 7), thus the greater the number of leads the more noise is introduced. More specifically, the F1-score, specificity and sensitivity of the 26 categories on 12-lead and 2-lead models are shown in table 8.

4. Discussion

Compared with our previous solution (Xu *et al* 2021), in this work we introduce several new techniques to handle the summarized challenges in ECG classification. Firstly, inspired by Han *et al* (2021), we evaluated the generalizability of the developed models with a dataset-wise cross-evaluation approach, which is not evaluation on the same distribution. The averaged cross-evaluation score is a good indicator of model performance on an unknown distribution. Secondly, in our previous solution we used a wavelet denoising module to filter the high-frequency noise. However, low-frequency noise also confused the classification and thus we used a bandpass filter to remove both high-frequency and low-frequency noise in this work. Thirdly, when we enlarged the

**Table 6.** Abbreviation of scored disease names.

Index	Disease name	Abbreviation
1	Atrial fibrillation	AF
2	Atrial flutter	AFL
3	Bundle branch block	BBB
4	Bradycardia	Brady
5	Complete left bundle branch block	CLBBB
6	Complete right bundle branch block	CRBBB
7	First degree atrioventricular block	IAVB
8	Incomplete right bundle branch block	IRBBB
9	Left axis deviation	LAD
10	Left anterior fascicular block	LAnFB
11	Left bundle branch block	LBBB
12	Low QRS voltages	LQSV
13	Nonspecific intraventricular conduction disorder	NSIVCB
14	Sinus rhythm	NSR
15	Premature atrial contraction	PAC
16	Pacing rhythm	PR
17	Poor R wave progression	PRWP
18	Premature ventricular contractions	PVC
19	Prolonged PR interval	LPR
20	Prolonged QT interval	LQT
21	Q wave abnormal	QAb
22	Right axis deviation	RAD
23	Right bundle branch block	RBBB
24	Sinus arrhythmia	SA
25	Sinus bradycardia	SB
26	Sinus tachycardia	STach
27	Supraventricular premature beats	SVPB
28	T wave abnormal	TAb
29	T wave inversion	TInv
30	Ventricular premature beats	VPB

kernel size in SE_Resnet models, the generalizability on an unknown distribution became more robust. Although the larger kernel size will increase the computational cost exponentially, we modified the kernel size from 7 to 9 to capture wider morphological features. Fourthly, during our experiment, the cross-relabeling strategy outperformed the consistent-distribution relabeling strategy used in the previous solution. We therefore applied the cross-relabeling strategy to derive high-confidence pseudo labels. Furthermore, we

Table 7. Labels in each datasets. Pseudo labels inferred via cross-relabeling are marked in bold.

	CPSC	CPSC_Extra	PTB_XL	Georgia	Chapman	Ningbo
AF	1221	170	1514	570	1780	6418
AFL	524	55	73	186	445	7615
BBB	1	21	27	116	16	385
Brady	340	282	6	6		7
CLBBB	9	1	40	33	62	213
CRBBB	1031	113	542	28	322	1096
IAVB	722	106	797	769	247	893
IRBBB	198	86	1118	407	8	246
LAD	104	26	5146	940	382	1163
LAnFB	10	2	1626	180	68	380
LBBB	236	38	536	231	205	35
LQRSV	2	1	182	374	249	794
NSIVCB	29	5	789	203	235	536
NSR	918	84	18 092	1752	1826	6299
PAC	616	76	398	639	258	1054
PR	17	3	296	113	20	1182
PRWP				1	3	638
PVC	170	188	494	161	300	1091
LPR			340	8	12	40
LQT	24	4	118	1391	57	337
QAb	4	1	548	464	235	828
RAD	39	1	343	83	215	638
RBBB	1857	3	577	542	454	195
SA	198	13	772	455	136	2550
SB	243	45	637	1677	3889	12 670
STach	305	315	826	1261	1568	5687
SVPB	11	57	157	1		9
TAb	150	22	2345	2306	1876	5167
TInv	18	6	294	812	157	2720
VPB		8	96	357	294	613

modified the sign loss function used in the previous solution with two augmentation parameters, where the augmentation strength could be adjusted freely.

Compared with other leading solutions (Han *et al* 2021, Nejedly *et al* 2021, Wickramasinghe and Athif 2021) in the Physionet/Computing in Cardiology Challenge 2021, we exclusively adopted (1) a different preprocessing workflow to mitigate data divergence, (2) SE modules with a large kernel size to capture wider channel-wise dependences for various lead versions, (3) the cross-relabeling strategy and the sign-augmented loss function to tackle the corrupted labels in training data and (4) a pos-if-any-pos ensemble strategy to make sufficient predictions. Class-specific threshold optimization makes a leading solution (Nejedly *et al* 2021) perform well on the challenge metric. In our experiment, we found that the performance of class-specific thresholds depended largely on how the signals in the unknown test datasets were partially labeled. We therefore used a single threshold for all the classes and our solution performed well on both challenge metric scores and AUC scores. Some works (Han *et al* 2021, Li *et al* 2021) adopted the asymmetric loss function to handle the missing labels. However, the asymmetric loss function will ignore negative classes lower than a hard threshold (Ridnik *et al* 2021), which derives a more sensitive prediction after threshold modification. The work combines time-domain input and frequency domain input (Wickramasinghe and Athif 2021), which is another compelling solution. In our work, we did not include the frequency domain input directly but modified the frequency domain information with bandpass filtering, which was effective for improving generalizability.

Previous work (Zhu *et al* 2021) combined a rule-based model and deep learning model for ECG classification. Theoretically, the rule-based model is a good choice for improving generalization performance. However, the accuracy of rule-based models is heavily dependent on the feature detection algorithm (like R-peak detection). In our experiment, although we tried to modify the parameters of some detection algorithms (Pan and Tompkins 1985) to cover various situations, the detection was not robust and the performance on an unseen distribution could not be guaranteed. Using other advanced methods (such as machine learning methods or deep learning methods) to detect traditional features is also biased on unseen distribution.

Another general choice in ECG classification are sequence model-based methods. Unlike CNN-based methods, which mainly focus on morphological features, sequence models, such as recurrent neural networks (Warrick and Homsy 2017, Singh *et al* 2018) and Transformers (Vaswani *et al* 2017, Natarajan *et al* 2020) are

Table 8. F1-score, specificity and sensitivity of each ECG abnormality on 12-lead and 2-lead models.

	12-lead			2-lead		
	F1	Specificity	Sensitivity	F1	Specificity	Sensitivity
AF	0.62	0.93	0.89	0.62	0.94	0.86
AFL	0.82	0.96	0.98	0.82	0.95	0.98
BBB	0.44	0.99	0.51	0.34	0.99	0.37
Brady	0.42	0.99	0.39	0.48	0.99	0.49
CLBBB/LBBB	0.78	0.99	0.89	0.79	0.99	0.87
CRBBB/RBBB	0.89	0.99	0.94	0.85	0.98	0.91
IABV	0.73	0.98	0.88	0.73	0.98	0.86
IRBBB	0.52	0.98	0.71	0.28	0.97	0.39
LAD	0.76	0.95	0.93	0.73	0.93	0.93
LAnFB	0.68	0.98	0.85	0.66	0.98	0.88
LPR	0.51	0.99	0.65	0.43	0.99	0.63
LQRSV	0.27	0.98	0.32	0.22	0.98	0.22
LQT	0.57	0.98	0.68	0.52	0.98	0.63
NSIVCB	0.40	0.98	0.46	0.38	0.98	0.46
NSR	0.93	0.94	0.97	0.93	0.93	0.97
PAC/SVPB	0.72	0.98	0.82	0.70	0.98	0.79
PR	0.90	0.99	0.91	0.91	0.99	0.92
PRWP	0.29	0.99	0.46	0.05	0.99	0.03
PVC/VPB	0.71	0.99	0.76	0.69	0.99	0.77
QAb	0.46	0.97	0.63	0.37	0.98	0.43
RAD	0.58	0.98	0.81	0.61	0.99	0.77
SA	0.76	0.98	0.86	0.75	0.98	0.84
SB	0.97	0.98	0.98	0.97	0.98	0.98
Stach	0.94	0.99	0.98	0.94	0.99	0.98
TAbs	0.59	0.82	0.87	0.56	0.79	0.85
TInv	0.47	0.93	0.70	0.43	0.93	0.63

good at capturing temporal dependences, especially when the input length is much longer and not fixed among different samples. In the future, we plan to combine the advantage of convolutional models in identifying morphological features and sequence models in capturing temporal dependences to perform ECG classification.

5. Conclusions

In this work, a deep learning framework is proposed for classification of ECG abnormalities. This study makes three main contributions. Firstly, we summarize the challenges of identifying ECG abnormalities and adopt a series of approaches to overcome them. Secondly, using the proposed framework, we train the models from multiple large datasets with sufficient labels, which can cover 30 types of ECG abnormalities and handle data with different lead combinations, while previous works focused on smaller datasets, more partial labels, fixed lead combinations and fewer abnormality types. Finally, we conduct experiments on both offline and online datasets to examine the proposed approaches, and the results on real-world datasets demonstrate the effectiveness of the proposed approaches. We believe that the models built by our proposed framework have the potential to be deployed in clinical practice.

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